Rethinking "Signal-To-Noise": A Coherent Beta-Binomial Reliability Formulation for Assessing Quality Measures

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Statements and Declarations

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suggestions provided by our many collaborators in the original measure development work.

Conflicts of interest/Competing interests

The authors have no conflicts of interest to declare that are relevant to the content of this article.

Author contributions

All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Samuel Field, Pat Malone, and Fei Dong. The first draft of the manuscript was written by Samuel Field and Fei Dong and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Data availability statement

The data that support the findings of this study are not openly available due to reasons of sensitivity of patient information. A modified version of the data that does not contain real patient information is available from the corresponding author upon reasonable request.

Abstract

Reliability assessment is a critical step for endorsing health quality measures used in provider profiling and pay-for-performance programs. A widely used "signal-to-noise" approach (Adams, 2009) for proportional (e.g., 0%-100%) measures is inconsistent with classical test theory (CTT) and can produce unstable estimates. This paper derives and advocates for an alternative beta-binomial reliability formulation that is mathematically equivalent to the Empirical Bayes (EB) shrinkage factor and fully aligns with CTT. We compare both methods using two real-world contraceptive care measures from Iowa Medicaid data: one with moderate rates ("Most-Mod") and one with low, near-boundary rates ("LARC"), aggregated for 99 counties. Results show that while both methods produce similar estimates for the moderate-rate measure, the Adams (2009) approach fails for the low-rate measure, yielding volatile and nonsensical estimates. The CTT-consistent formulation provides stable and interpretable results across all data.

Keywords. Performance quality measure, reliability, beta-binomial model, Bayesian, healthcare quality measurement

Introduction

Performance quality measures used in public reporting, pay-for-reporting, or pay-for performance programs require validity and reliability evidence to obtain scientific endorsement (Partnership for Quality Measures 2025). A quality measure is *valid* to the extent that it measures what is intended; and *reliable* to the extent that it provides

consistent results across repeated measurements, or to the extent it can dependably differentiate levels of performance among units being measured (Lord and Novick 1968).

A key question for quality improvement is whether one can consistently classify high or low performance among health care providers (e.g., physician, clinic, hospital). Reliable measurement reduces the chance of misclassifying a provider's performance, and therefore allows for better targeting of quality improvement resources.

We focus on the common situation of binary data proportions (e.g., outcome success, service provision, testing completed, etc.). In the health quality field, proportions are often reported as rates or percentages of successful outcomes per patient within a timeframe, ranging from 0 (0%) to 1 (100%). One widely used method to assess reliability for observed proportions was originally proposed by Adams (2009) in a technical report from the RAND Corporation. The National Quality Forum (NQF) sanctioned this method as an acceptable reliability reporting approach for measure developers. The approach was informally described as a 'signal-to-noise' analysis and employed a beta-binomial model.

Although reliability evidence is required for quality measure endorsement, we agree with other authors (Nieser and Harris 2024) that the current approaches to estimating reliability often make strong assumptions about the data that do not align with typical requirements from measurement theory (Kuder and Richardson 1937; Lord and Novick 1968; He et al. 2019). Although this point was made briefly in a methodological review of over 14 different reliability calculations based on observed proportions (Nieser and Harris 2024), we want to heighten the awareness of the issue. The characteristics of real-world health services data do not easily fit models for reliability estimation. One fundamental

challenge is the difficulty of making repeated measurements under substantially similar circumstances. Consider that each provider operates from different clinical contexts in the same network; each patient presents a different clinical history; the number and types of patients per provider varies widely, and every geographical facility provides services to a unique population under different policies and resource constraints. Reliability estimates derived from these observed data use simplifying assumptions that do not account for the many influences affecting performance among entities (e.g., providers, facilities, etc.). A fuller discussion of the various approaches and assumptions for reliability estimation can be found in Nieser and Harris (2024).

The broader question of how closely these approaches can approximate standard concepts of measurement reliability when using extant health measures data deserves further work and is beyond the scope of this paper. However, given that reliability testing is a *de facto* prerequisite for measure endorsement, we discuss and derive a simpler and more consistent approach first identified by Nieser and Harris (2024) and recommend it as a superior alternative to the widely used Adams (2009) (aka 'signal-to-noise') approach. Despite early support from NQF for the Adams (2009) formulation and its subsequent widespread usage in empirical studies of health care quality (Adams and Paddock 2017; Blair et al. 2015; Kazis et al. 2017; Staggs and Cramer 2016), the approach is not consistent with classical measurement theory.

Our manuscript has two objectives. First, we derive the alternative reliability measure identified by Nieser and Harris (2024) and equate it to the reliability parameter used for Empirical Bayes (EB) shrinkage. We argue that this formulation is consistent with the

commonly accepted definition of reliability. Second, we compare the proposed formulation to the Adams (2009) version using two health service quality measures. We also demonstrate that the proposed formulation functions better in situations with small numbers of patients per provider, as well as among providers with very-low or very-high rates. In Online Resource, we demonstrate a straightforward implementation of this reliability method using the R statistical software package.

Methods

The Beta-Binomial model for binary data

The measures of service quality we examine take the form of a binomial proportion of 'success':

$$\frac{y_i}{n_i}, \tag{1}$$

where y_i is a count of patients who were successfully provided a particular service in cluster i (e.g., counseling, treatment, etc.); and n_i is the total number of patients in the cluster who were eligible to receive that service. We can regard the sample proportions as an unbiased estimator of the underlying service probability, π_i

$$\hat{\pi}_i = \frac{y_i}{n_i}.$$

Our approach to modeling assumes that the observed counts of services received by patients within cluster i arise as a binomial random variable with parameters π_i and n_i:

$$p(Y = y_i | n_i, \pi_i) = Binomial(n_i, \pi_i). \tag{2}$$

Let us further assume that the cluster-level proportion parameters π_i, or "true" rates for providers, are sampled from a population of measure rates across clusters that follow a beta distribution with parameters α_0 and β_0:

$$p(\Pi = \pi_i | \alpha_0, \beta_0) = Beta(\alpha_0, \beta_0)\ . \tag{3}$$

When a parameter of a random variable is itself a random variable, the statistical distribution of that parameter is a prior distribution. In this case, the beta distribution is a prior distribution for the parameter π_i in the binomial distribution. Furthermore, by integrating over the uncertainty in π_i, one obtains the marginal distribution of the observed counts y_i, or the beta-binomial distribution:

$$p(Y = y_i | n_i, \alpha_0, \beta_0) = BetaBinomial(n_i, \alpha_0, \beta_0)\ . \tag{4}$$

In this case, the Beta distribution is a conjugate prior for the Binomial likelihood, and the resulting beta-binomial compound distribution is mathematically tractable and widely used in modeling binomial count data sampled from a diverse population of clusters (i.e. overdispersion).

In a fully Bayesian analysis, values for the prior distribution parameters α_0 and β_0 would be chosen in a manner that reflects knowledge or "beliefs" about the underlying distribution of π_i that is external to the collected data. Given the lack of any initial knowledge about π_i, we employ an EB approach, where prior distribution parameter values are estimated from the sample proportions as $\hat{\alpha}_0$ and $\hat{\beta}_0$.

As Adams (2009) notes, the beta distribution provides flexibility for fitting to observed distributions of service quality measures. Various combinations of the α_0 and β_0

parameters generate a wide range of shapes, including extreme U-shaped distributions in which high- and low-quality providers are grouped near each end of the distribution (Liu et al. 2013). Since this flexibility only requires the estimation of two parameters, the risk of over-fitting the observed data is minimal in cases where the number of clusters is large (e.g., > 30).

The posterior distribution for quality measures

From a Bayesian perspective, predictions regarding the "true" cluster-level service quality score for any given cluster (e.g., provider or county) are based on the posterior distribution of π_i. The posterior distribution is the distribution of possible values for π_i for each cluster, conditional on the observed measure rate in that cluster, $\frac{y_i}{n_i}$, as well as the estimated parameters in the prior beta distribution (α_0, β_0). Because of the conjugacy property discussed above, the posterior distribution is analytically tractable. The posterior distribution for $\Pi = \pi_i$ is proportional to another beta distribution:

$$p(\Pi = \pi_i | n_i, y_i, \alpha_0, \beta_0) \propto Beta((\alpha_0 + y_i), (\beta_0 + (n_i - y_i)) \,. \tag{5}$$

In the current context, the posterior distribution is the distribution of possible "true" cluster-level measure rates based on sampling a single quality measure (i.e., $\frac{y_i}{n_i}$) from a population of true measure rates that are assumed to follow a beta distribution with known or estimated parameters α_0 and β_0.

To derive reliability measures, we focus on the mean, $E(\pi_i)$, of this distribution:

$$E(\pi_i | n_i, y_i, \alpha_0, \beta_0) = E(Beta((\alpha_0 + y_i), (\beta_0 + (n_i - y_i))) \,. \tag{6}$$

We can substitute the analytically derived mean of the beta distribution, $\frac{\alpha_0}{(\alpha_o+\beta_0)}$, in the right side of the equation and simplify the result:

$$E(\pi_i|n_i, y_i, \alpha_0, \beta_0) = \frac{(\alpha_0+y_i)}{(\alpha_0+y_0)+(\beta_0+(n_i-y_i))} = \frac{(\alpha_0+y_i)}{(\alpha_0+\beta_0+n_i)} \ . \tag{7}$$

Equation (7) above provides the basis for a cluster-level estimate of service quality that is a mixture of observed data ($\boldsymbol{n_i}, \boldsymbol{y_i}$) and population parameters ($\boldsymbol{\alpha_0}, \boldsymbol{\beta_0}$) or estimates of the population parameters ($\widehat{\boldsymbol{\alpha}}_{\boldsymbol{0}}, \widehat{\boldsymbol{\beta}}_{\boldsymbol{0}}$). In the section below, we demonstrate how equation 7 can be used to derive an alternative to the Adams (2019) reliability measures for the sample proportions, $\widehat{\boldsymbol{\pi}}_{\boldsymbol{i}}$.

The empirical Bayes shrinkage estimator

Our derivation of the cluster-specific reliability estimate, λ_i, employs a commonly used identity for the EB shrinkage estimator. Substituting empirically-estimated values $\hat{\alpha}_0$ and $\hat{\beta}_0$ for unobserved population parameters α_0 and β_0, the mean of the posterior distribution is a combination of the observed proportions, $\frac{y_i}{n_i}$, and the mean of the prior beta distribution for the "true" measure rate, π_i, weighted by the reliability λ_i :

$$E(\pi_i|n_i, y_i, \hat{\alpha}_0, \hat{\beta}_0) = \hat{\pi}_{EB,i} = \frac{y_i+\hat{\alpha}_0}{n_i+\hat{\alpha}_0+\hat{\beta}_0} = \lambda_i\left(\frac{y_i}{n_i}\right) + (1-\lambda_i)\left(\frac{\hat{\alpha}_0}{(\hat{\alpha}_0+\hat{\beta}_0)}\right). \tag{8}$$

As the reliability λ_i approaches 1 for a given cluster (nearing perfect reliability), the mean of the posterior distribution for that cluster approaches the observed proportion. Conversely, for λ_i values less than 1, the mean of the posterior is "shrunk" towards the

(estimated) mean of the prior distribution $\frac{\hat{\alpha}_0}{\left(\hat{\alpha}_0+\hat{\beta}_0\right)}$. For any given cluster, the more the shrinkage estimator is pulled towards the mean of the prior distribution, the less reliable the observed quality measures are for that cluster.

Deriving the reliability formulation from the classical definition of reliability

As we show below, the reliability of the sample proportions can be obtained by solving for λ_i in equation (8), classical test theory (Novick 1965) can be used to obtain the same result in a straight-forward manner. In classical test score theory, reliability is defined as a ratio of true score variance to observed score variance:

$$\lambda_i = \frac{\sigma^2_{true}}{\sigma^2_{obs}} = \frac{\mathrm{var}(\pi_i)}{\mathrm{var}(\hat{\pi}_i)} \ . \tag{9}$$

In the beta-binomial model with EB estimation, the variance of the true score in the numerator is equal to the variance of the prior distribution for π_i as estimated from the data. The variance of a beta distributed random variable is:

$$\sigma^2_{true} = V\mathrm{ar}(\pi_i) = \frac{(\hat{\alpha}_0\hat{\beta}_0)}{(\hat{\alpha}_0+\hat{\beta}_0)^2*(\hat{\alpha}_0+\hat{\beta}_0+1)} \ . \tag{10}$$

The variance in the observed service incidence, y_i, is the variance of the marginal distribution for the observed counts, λ_i , *viz*. the beta-binomial distribution. However, we need to derive the variance of the observed proportions, $\frac{y_i}{n_i}$, rather than the variance of the observed counts, y_i. First, we note that the variance of any random variable multiplied by a constant is the variance of the random variable times the square of the constant. Thus,

$$\sigma_{obs}^2 = var\left(\frac{y_i}{n_i}\right) = var(y_i) \times \left(\frac{1}{n_i}\right)^2 \ . \tag{11}$$

Then we replace $var(y_i)$ with the variance of beta-binomial distribution. Thus,

$$\sigma_{obs}^2 = var(y_i) \times \left(\frac{1}{n_i}\right)^2 = \left(\frac{n_i(\alpha_0\beta_0)(\alpha_0+\beta_0+n_i)}{(\alpha_0+\beta_0)^2(\alpha_0+\beta_0+1)}\right) \times \left(\frac{1}{n_i}\right)^2 . \tag{12}$$

In the final step, the ratio of true score to observed score variance in the beta-binomial model, λ_i becomes:

$$\lambda_i = \frac{\sigma_{true}^2}{\sigma_{obs}^2} = \frac{(\alpha_0\beta_0)}{(\alpha_0+\beta_0)^2*(\alpha_0+\beta_0+1)} \Bigg/ \left(\frac{n_i(\alpha_0\beta_0)(\alpha_0+\beta_0+n_i)}{(\alpha_0+\beta_0)^2(\alpha_0+\beta_0+1)}\right) \times \left(\frac{1}{n_i}\right)^2 \ . \tag{13}$$

To simplify this expression, first let

$$A = \frac{(\alpha_0\beta_0)}{(\alpha_0+\beta_0)^2*(\alpha_0+\beta_0+1)} \ .$$

Now,

$$\lambda_i = \frac{A}{A\cdot\left(\frac{n_i(\alpha_0+\beta_0+n_i)}{n_i^2}\right)} = \frac{1}{\left(\frac{n_i(\alpha_0+\beta_0+n_i)}{n_i^2}\right)} = \frac{n_i^2}{n_i(\alpha_0+\beta_0+n_i)},$$

which simplifies to,

$$\lambda_i = \frac{n_i}{\alpha_0+\beta_0+n_i} \ . \tag{14}$$

We can then substitute $\frac{n_i}{\hat{\boldsymbol{\alpha}}_0+\hat{\beta}_0+n_i}$ for λ_i in equation (8).

$$E\left(\pi_i \middle| n_i, y_i, \hat{\alpha}_0, \hat{\beta}_0\right) = \left(\frac{n_i}{\hat{\alpha}_0+\hat{\beta}_0+n_i}\right)\left(\frac{y_i}{n_i}\right) + \left(1 - \left(\frac{n_i}{\hat{\alpha}_0+\hat{\beta}_0+n_i}\right)\right)\left(\frac{\hat{\alpha}_0}{(\hat{\alpha}_0+\hat{\beta}_0)}\right). \tag{15}$$

Now we simplify and rearrange both terms:

- First term:
  - $\left(\frac{n_i}{\hat{\alpha}_0+\hat{\beta}_0+n_i}\right)\left(\frac{y_i}{n_i}\right) = \frac{y_i}{n_i+\hat{\alpha}_0+\hat{\beta}_0}$.
- Second term:
  - $\left(1-\left(\frac{n_i}{\hat{\alpha}_0+\hat{\beta}_0+n_i}\right)\right)\left(\frac{\hat{\alpha}_0}{(\hat{\alpha}_0+\hat{\beta}_0)}\right) = \left(\frac{\hat{\alpha}_0+\hat{\beta}_0}{n_i+\hat{\alpha}_0+\hat{\beta}_0}\right)\left(\frac{\hat{\alpha}_0}{(\hat{\alpha}_0+\hat{\beta}_0)}\right) = \frac{\hat{\alpha}_0}{n_i+\hat{\alpha}_0+\hat{\beta}_0}$.

By combining these two terms we confirm the identity of the EB estimator:

$$E(\pi_i|n_i, y_i, \alpha_0, \beta_0) = \frac{y_i}{n_i+\hat{\alpha}_0+\hat{\beta}_0} + \frac{\hat{\alpha}_0}{n_i+\hat{\alpha}_0+\hat{\beta}_0} = \frac{y_i+\hat{\alpha}_0}{n_i+\hat{\alpha}_0+\hat{\beta}_0} \quad .$$

The Carlin and Louis (2000) text on Bayesian methods contains an equivalent expression for the beta-binomial reliability parameter (pages 67-68), and to our knowledge it has only recently appeared in the applied statistics literature; Nieser and Harris (2024) presented this finding while also noting its absence in the published literature. Their derivation was based on decomposing σ^2_{obs} into two additive components based on the law of total variance. The law of total variance in this context breaks down the distribution of the observed proportions into two components:

$$\sigma^2_{obs} = var\left(\frac{y_i}{n_i}\right) = var(\hat{\pi}_i) = var(E[\hat{\pi}_i|\pi_i]) + E[var(\hat{\pi}_i|\pi_i)] \quad . \qquad (16)$$

In words, the marginal variance of the observed proportions as estimators of the underlying service probability is the variance of the expected value of the estimator conditional on the true value plus the conditional variance of the estimator averaged or integrated over the PD of π_i (i.e. the average measurement error). The first term is straightforward. Since $\hat{\pi}_i$is an unbiased estimator of π_i, we can write,

$$var(E[\hat{\pi}_i|\pi_i]) = var(\pi_i) = \frac{(\alpha_0\beta_0)}{(\alpha_0+\beta_0)^2(\alpha_0+\beta_0+1)} \ .$$

Now reliability becomes:

$$\lambda_i = \frac{var(\pi_i)}{(var(\pi_i) + E[var(\hat{\pi}_i|\pi_i)])} \tag{17}$$

Nieser and Harris (2024) derive an expression for the binomial variance, averaging over all possible values for π_i (see Appendix A.3). However, a simple rearrangement of equation (17) yields:

$$\frac{1}{\lambda_i} = \frac{(var(\pi_i) + E[var(\hat{\pi}_i|\pi_i)])}{var(\pi_i)} \ ,$$

$$E[var(\hat{\pi}_i|\pi_i)] = \frac{var(\pi_i)}{\lambda_i} - var(\pi_i) \ ,$$

$$E[var(\hat{\pi}_i|\pi_i)] = \frac{\frac{(\alpha_0\beta_0)}{(\alpha_0+\beta_0)^2(\alpha_0+\beta_0+1)}}{\frac{n_i}{\alpha_0+\beta_0+n_i}} - \frac{(\alpha_0\beta_0)}{(\alpha_0+\beta_0)^2(\alpha_0+\beta_0+1)} \ ,$$

$$E[var(\hat{\pi}_i|\pi_i)] = \frac{(\alpha_0\beta_0)}{n_i(\alpha_0+\beta_0)(\alpha_0+\beta_0+1)} \ . \tag{18}$$

Revisiting the Adams (2009) formulation of reliability

We showcase the Nieser and Harris (2024) derivation above because it resembles the "signal-to-noise" approach employed by Adams (2019). Adams similarly defines $\mathrm{var}(\hat{\pi}_i)$ as a sum of two components, yet replaces the last term in equation (14), $E[var(\hat{\pi}_i|\pi_i)]$, with $var(\hat{\pi}_i|\pi_i = \hat{\pi}_i)$, yielding:

$$\mathrm{var}(\hat{\pi}_i) = var(E[\hat{\pi}_i|\pi_i]) + var(\hat{\pi}_i|\pi_i = \hat{\pi}_i). \tag{19}$$

Instead of averaging the conditional variance over all possible values of π_i where $\pi_i \sim Beta(\alpha_0, \beta_0)$, Adams decides to use the conditional error variance of the sample proportions for $\pi_i = \hat{\pi}_i = \frac{y_i}{n_i}$ where $y_i \sim Binomial(\pi_i, n_i)$, leading to:

$$var(\hat{\pi}_i | \pi_i = \hat{\pi}_i) = \left(\frac{1}{n_i}\right)^2 n_i \hat{\pi}_i \times (1 - \hat{\pi}_i) = \frac{\hat{\pi}_i \times (1 - \hat{\pi}_i)}{n_i}. \tag{20}$$

After substituting the expression for the variance of the Beta distribution, he provides his formulation of reliability:

$$\lambda_i = \frac{\left(\frac{(\alpha_0 \beta_0)}{(\alpha_0 + \beta_0)^2 * (\alpha_0 + \beta_0 + 1)}\right)}{\left(\frac{(\alpha_0 \beta_0)}{(\alpha_0 + \beta_0)^2 * (\alpha_0 + \beta_0 + 1)} + \frac{\hat{\pi}_i \times (1 - \hat{\pi}_i)}{n_i}\right)} \ . \tag{21}$$

And with simplification, we also obtain the version of the Adams formulation as reported in Nieser and Harris (2024).

$$\lambda_i = \frac{\alpha_0 \beta_0}{\alpha_0 \beta_0 + (\alpha_0 + \beta_0)^2 * (\alpha_0 + \beta_0 + 1) \frac{\hat{\pi}_i \times (1 - \hat{\pi}_i)}{n_i}} .$$

Now we can see differences between the Adams formulation of beta-binomial reliability and the one derived from classical test theory. Adams (2009) chose to use

$$var(\hat{\pi}_i | \pi_i = \hat{\pi}_i) = \frac{\hat{\pi}_i \times (1 - \hat{\pi}_i)}{n_i}$$

as the measurement error variance component of the observed proportions, $\hat{\pi}_i$, whereas the classical definition uses:

$$E[var(\hat{\pi}_i | \pi_i)] = \frac{(\alpha_0 \beta_0)}{n_i (\alpha_0 + \beta_0)(\alpha_0 + \beta_0 + 1)} \ .$$

The classical test theory definition of reliability has a clear interpretation and utility since the variance formulations used for the true and observed scores are both global measures of variation. This feature does not hold for the Adams approach, which appears to be a ratio of global "signal", $Var(\pi_i)$, and local "noise", $Var(\hat{\pi}_i|\pi_i = \hat{\pi}_i)$, with a misleading or incoherent interpretation. Nieser and Harris also make the same point (see page 4582).

Example performance measures and data source

To demonstrate the methods discussed in this article, we use two performance measures for contraceptive care developed by the U.S. Department of Health and Human Services Office of Population Affairs: the "most or moderately effective contraceptive methods" (Most-Mod) measure and the "long-acting reversible contraceptive methods" (LARC) measure (Gavin et al. 2017). These measures were endorsed by the NQF in 2021 for evaluating access to contraceptive services using administrative claims data. The measures are defined as the proportion of all eligible reproductive age (15 – 44 year) women who are using a Most-Mod method (Most-Mod measure) or provided a LARC method (LARC measure) during a measurement period. We used deidentified, aggregated claims data from Medicaid beneficiaries residing in Iowa in 2018. The Most-Mod and LARC measures were calculated at the county level for the 99 counties in Iowa (included in Online Resource Appendix A). The methodology provided is generalizable to other quality measures that are proportional by design, and based on claims, electronic health records (EHR), or other forms of health outcome datasets.

Results

To illustrate the EB analysis, we plot the density of the beta distribution with maximum likelihood estimates (MLE) of α_0 and β_0 obtained from a beta-binomial regression of the Most-Mod measure from 99 counties in Iowa. Figure 1 depicts the distribution of observed quality measures, $\hat{\pi}_i$, as a histogram, while the fitted beta distribution of the underlying true proportions, π_i, is depicted as a continuous density plot. The mean, median, and mode of the distribution is close to 0.4, indicating that in the average county, approximately 40% of women used a Most-Mod method during the measurement year. In addition, most of the county-level proportions fell within an interval of 0.2 to 0.6. As can be seen from the plot, the fit to the beta distribution is approximate. The fitted distribution does not capture a small cluster of counties with observed proportions, $\hat{\pi}_i > 0.6$. As an approximation, however, this true score distribution does appear to fit the observed score distribution quite well.

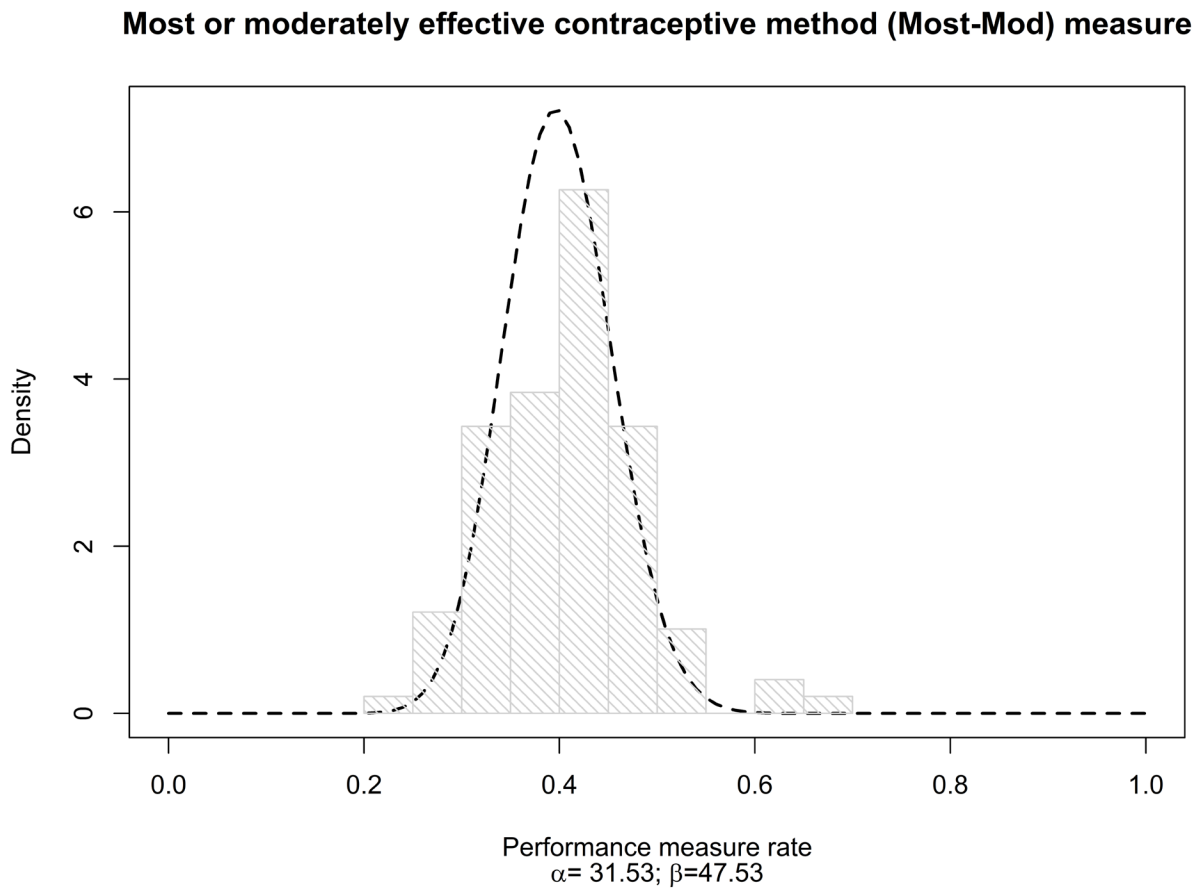

Fig. 1 *Histogram with beta distribution density plot (dotted line) overlay depicting the county-level distribution of the most or moderately effective contraceptive methods (Most-Mod) measure among 99 Iowa counties in 2018*

Although not evident in Figure 1, the range (i.e., x-axis) of the beta distribution is restricted to the 0 to 1 interval, making it a particularly suitable prior for a binomial parameter, π_i. In Figure 2, we present a similar density plot for the LARC measure, which has a much lower measure rate. In contrast to the estimated prior distribution for the Most-Mod measure, the distribution of the LARC measure is shifted considerably to the left with a noticeable right skew.

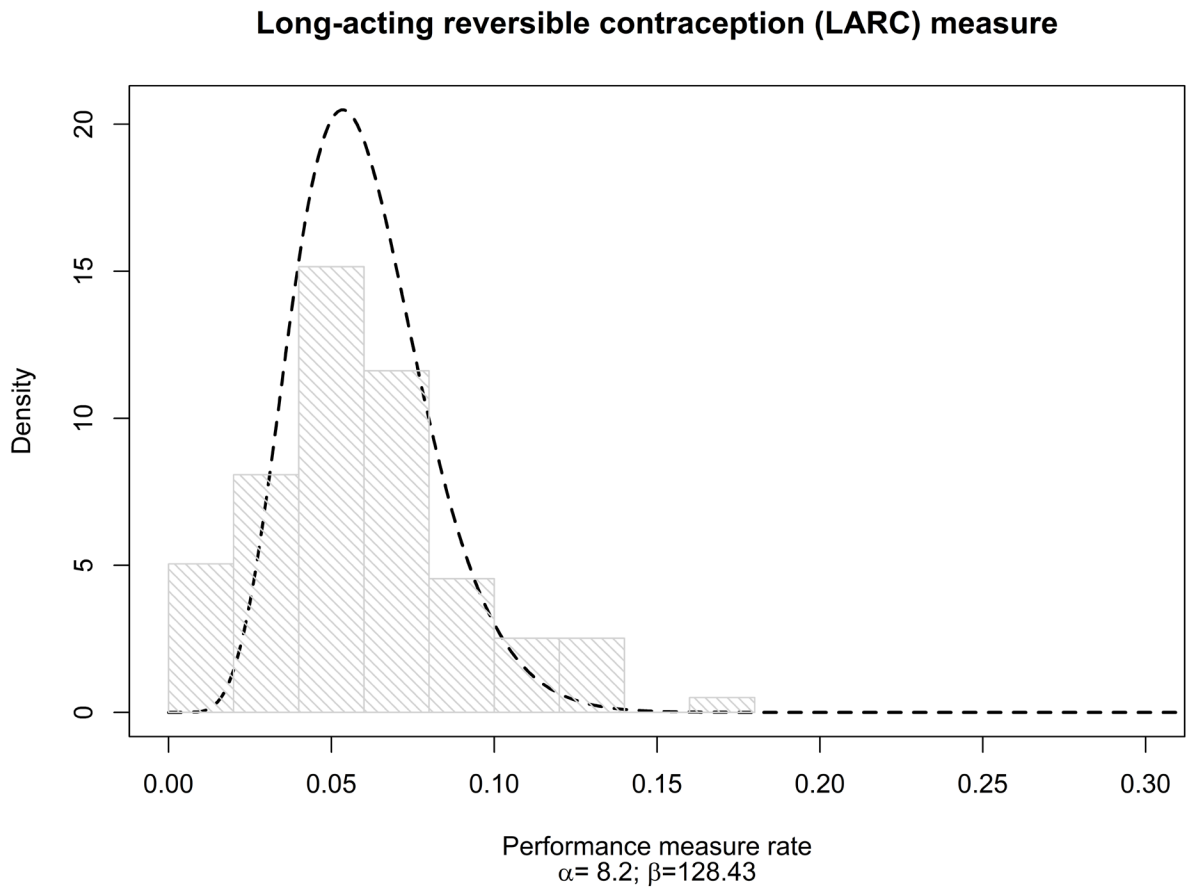

Fig. 2 *Histogram with beta distribution density plot (dotted line) overlay depicting the county-level distribution of the long-acting reversible contraceptive methods (LARC) measure among 99 Iowa counties in 2018*

In Figure 3, we illustrate the Most-Mod measure posterior distributions for two different counties are plotted on top of the fitted prior distribution previously seen in Figure 1. We also indicate the means of each posterior distribution as dotted vertical lines; and of the observed measure $\left(\frac{y_i}{n_i}\right)$ as thick, dashed vertical lines. The arrows indicate the direction and magnitude of the shrinkage of the EB estimate towards the mean of the prior distribution.

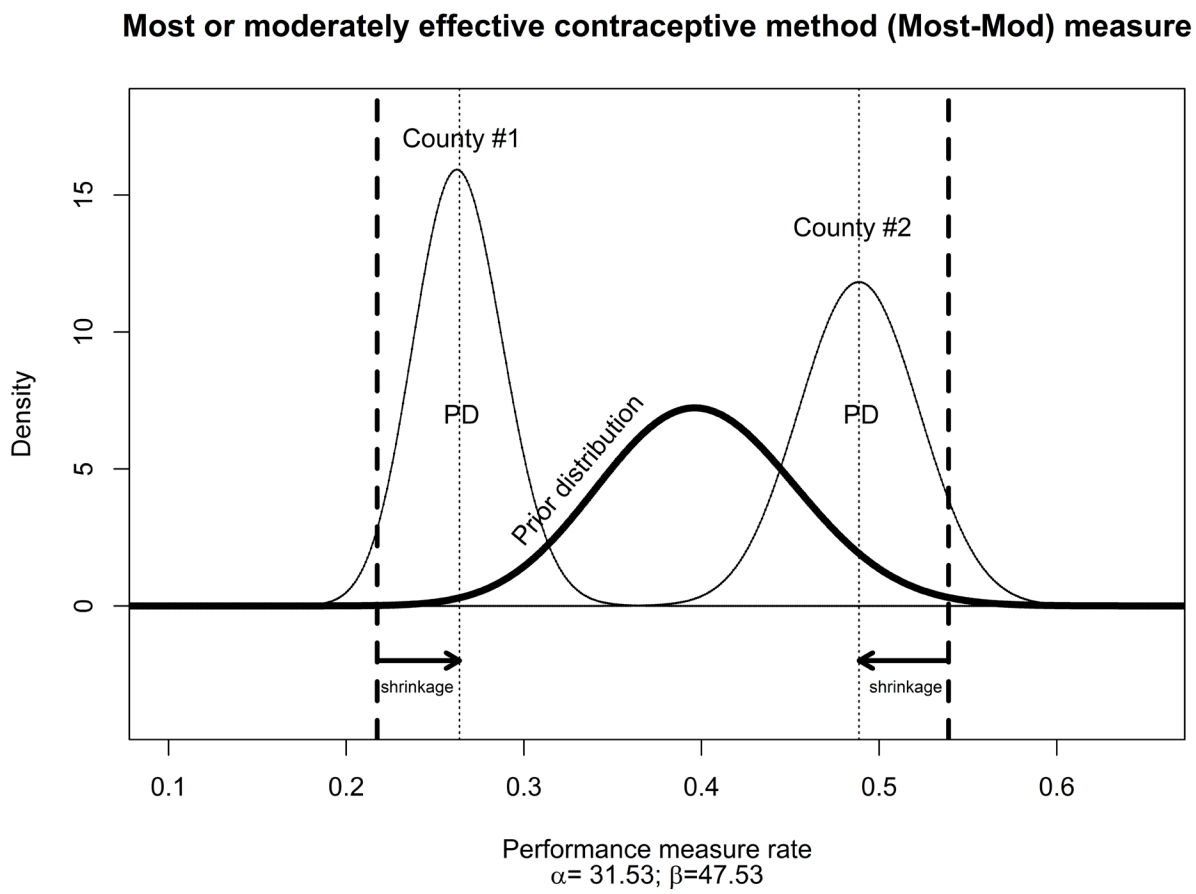

Fig. 3 *Posterior distribution (PD) for two Iowa counties with a prior distribution overlay: most or moderately effective contraceptive methods (Most-Mod) measure*

Although the general shapes of the plotted posterior distributions are similar, the more peaked distribution seen for the county on the left (county #1) reflects greater certainty about the location of that county's "true" measure rate. As our formulation of reliability (14) indicates, this difference between the two counties is entirely a function of their difference in size. The numbers of patients in counties #1 and #2 are 230 and 141, respectively.

To demonstrate the practical consequences of the Adams (2019) formulation versus our proposed alternative to calculating beta-binomial reliability, we return to the 99-county example for the two quality measures of Most-Mod and LARC. In Figure 4, we plot county-level reliability parameters calculated using the proposed formulation (14) against the results obtained under the Adams (2009) approach (18). The proposed reliability calculations depend exclusively on the cluster size (i.e., total number of women) while the Adams approach depends on both the cluster size and the observed incident rate of the cluster. Thus, in comparing the two measures obtained from the same population of clinics, it is useful to distinguish clinics by their size and incident rate. We do this in the figures below; the underlying data on which the plots were based can be found in Online Resource Appendix A which directly compares the county-level reliability estimates. The estimates were calculated using the standard statistical software R (R version 4.3.2, http://www.r-project.org), and the syntax is provided in Online Resource Appendix B.

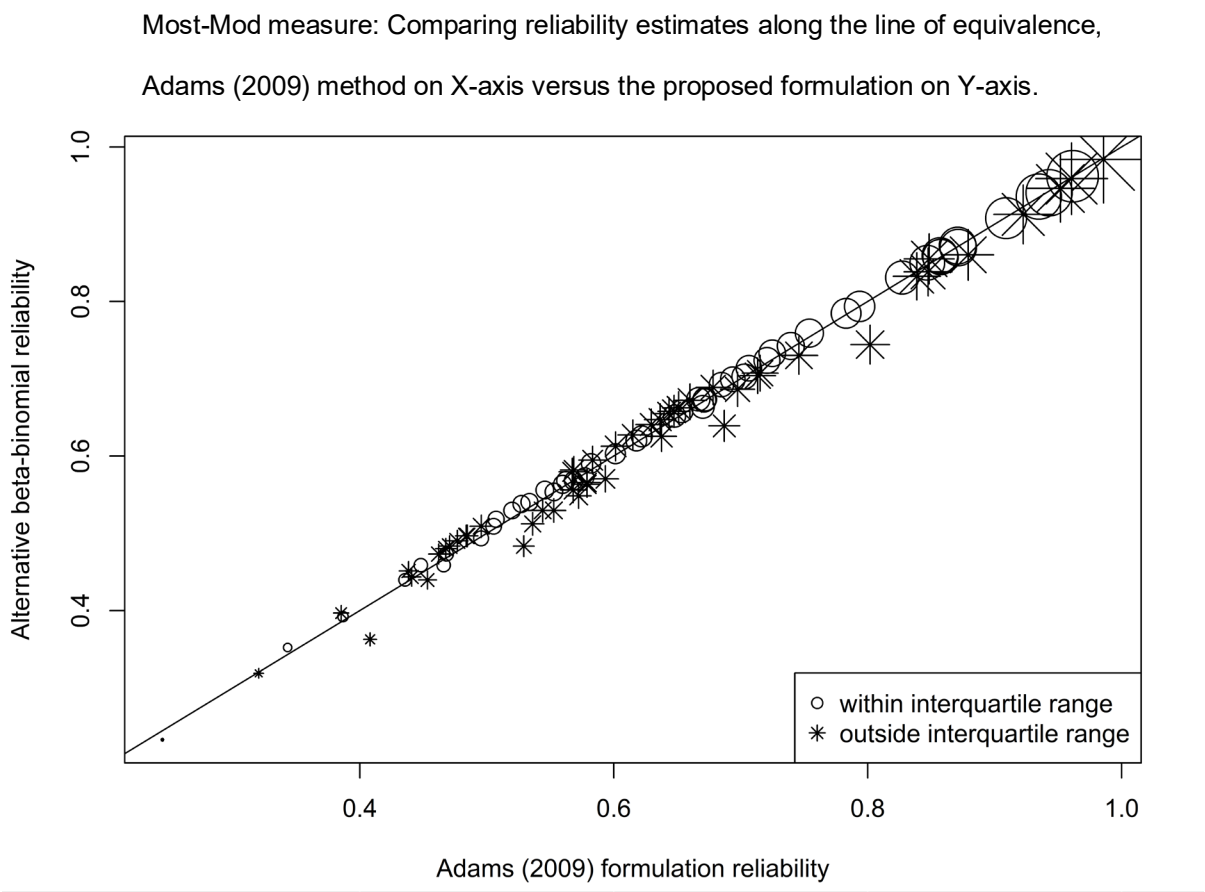

Fig. 4 *Comparison of reliability parameters between our alternative formulation and the Adams (2009) formulation: most or moderately effective contraceptive methods (Most-Mod) measure among 99 Iowa counties in 2018. The diagonal line represents the line of equality between the two methods of estimation, and each data point represents the value of the two estimates for each county. The size of the data points are proportional to the natural log of the cluster size, ln(n_i), and the plotting character identifies counties with observed measures (y_i/n_i) that are either within (circle) or outside (asterisk) the interquartile range of the sample*

Figure 4 reveals few differences between these two approaches, and as expected smaller counties with fewer patients tend toward lower reliability. However, when we turn to the

same plot for the LARC measure shown in Figure 5, substantial differences between the two methodologies emerge. Specifically, the disagreement between the methods appears greater in counties where the observed measure lies outside versus inside the inter-quartile range of the sample.

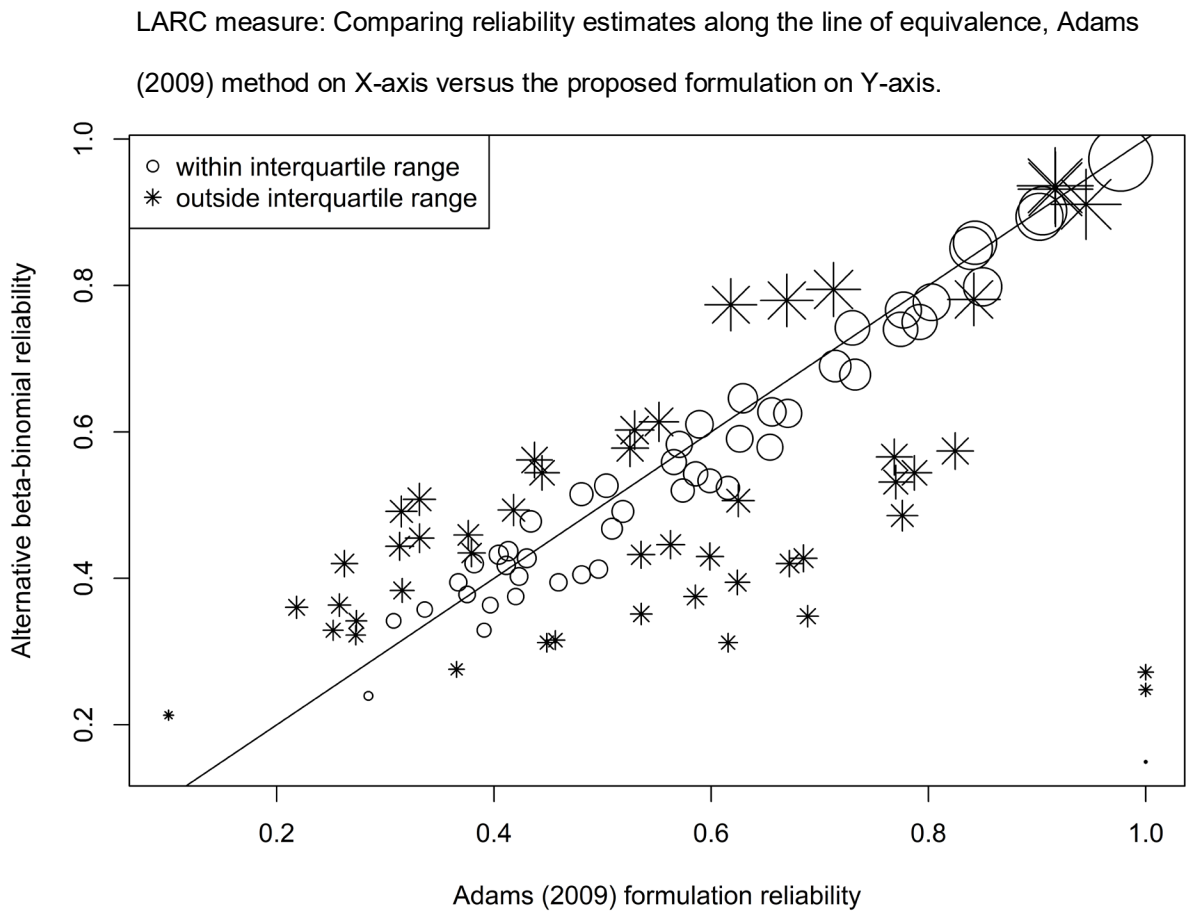

Fig. 5 *Comparison of reliability parameters between our alternative formulation and* the Adams (2009) *formulation: long-acting reversible contraceptive methods (LARC) measure among 99 Iowa counties in 2018*

This pattern of results is expected because the reliability calculations that we describe depend *exclusively* on the cluster size (i.e., total number of women) while the Adams approach depends on both the cluster size and the observed rate of the cluster (i.e., women

who received service in the measurement year). Differences were not as evident for the Most-Mod rates since these mostly occurred near the middle of the 0 to 1 interval where binomial variance is relatively insensitive to changes in the binomial parameter ρ. But LARC rates are much lower than Most-Mod rates. As cluster-level rates move toward the boundaries, the between-cluster differences in the binomial variance component of the reliability parameter become much larger, so identically sized clusters can produce dramatically different reliabilities within the Adams formulation. In the proposed formulation, identically sized clusters sampled from the same prior distribution will have equal reliability regardless of cluster rates.

A particularly instructive example can be found in the cluster of five moderately-sized counties shown as asterisks in Figure 5 toward the right and below the line of equivalence. Under the Adams (2009) formulation, these counties have reliability estimates around 0.80, yet within our proposed formulation, have much lower estimates in the 0.40-0.50 range. Reliability is higher under the Adams approach because the observed LARC provision rate is considerably lower than the average. As a result, the local estimate of binomial variance employed by Adams, $var(\hat{\pi}_i|\pi_i = \hat{\pi}_i)$ is also *lower* than the binomial variance used in our recommend approach. Our estimate of variance instead averages over the beta-distributed "true" rate $E[var(\hat{\pi}_i|\pi_i)]$. As we noted previously, the Adams formulation is inconsistent with a classical definition of reliability by mixing global signal (beta variance) with local noise (binomial variance). We are not arguing that our approach is preferred due to consistency with the beta-binomial model; rather that our approach is derived from a more coherent and widely accepted statistical framework.

It is also worth noting that the three small counties that appear in the bottom-right corner of Figure 5 had *perfect* (1.0) reliability under the Adams approach, a nonsensical result; but properly showed poor reliability using our formulation. This is because within the Adams approach, variance of the binomial distribution equals zero when the observed measure is either 0 or 1.

Discussion

Our derivation of the proposed formulation is consistent with the classical concept of reliability as the proportion of true score variance to total measurement variance. As Nieser and Harris (2024) note, this formulation can also be interpreted as the correlation among service proportions that are repeatedly sampled from a constant, provider-specific service probability (π_i); or as the squared correlation between the observed proportions and the true proportions over all providers. We further show the application of reliability as the shrinkage factor in EB estimation.

In practice the Adams formulation should yield reliability estimates similar to classical reliability estimates when there are large numbers of patients per provider and rates between 10-90%; however the Adams method will diverge as numbers of patients per provider become smaller, or when rates approach the boundaries of 0% or 100%.

Measurement strategies that involve sampling more patients per clinic will result in higher reliability when applied to the same population. Despite this reasonable application of classical test theory, Adams (2009) writes,

> Higher reliability increases the likelihood that you will assign a physician to the "right" group. Sample size, while often used as a proxy for reliability, may be insufficient. Therefore, simple minimum sample size cut points (e.g., a rule that a physician's profile must include at least 30 events) may not solve this problem.

We do not agree. The practice of using reliability to set a minimum sample size for reporting purposes is both a reasonable and a conventional application of classical test theory. In another section, Adams (2009) writes that "[reliability] is often mistakenly thought of as a property of a measurement system (e.g. the SF-12 survey)". In this limited situation, where a single binomial count is available from each provider, the within-provider sample size is better thought of as an *uncontrolled* property of the measurement system in the same way that a psychometrician would regard the number of items (e.g. survey questions) used in the construction of a summative index.

Conclusion

In this paper, we focused on the commonly used "signal-to-noise" approach to beta-binomial reliability developed by Adams (2009). Following Nieser and Harris (2024) we derive an alternate method that aligns with classical test theory, further explicate an error made by Adams, and examine the practical consequences of this error in a real-world setting. Although the Adams formulation has become accepted practice and was applied in numerous empirical studies of healthcare quality (Adams and Paddock 2017; Blair et al. 2015; Kazis et al. 2017; Staggs and Cramer 2016), the resulting estimates are inconsistent with standard measurement theory. We therefore recommend the simpler and more

coherent formulation shown in (17) and (18), or the permutation split-sample method as described by Nieser and Harris (2024).

We also note that the standard definition of reliability is more appropriately seen as a property of a measurement instrument rather than as a property of the unit being measured. In populations of providers where the within-provider sample size is not controlled, we might suggest using reliability analysis to establish sample-size based cut-offs; researchers can then report overall reliability as a minimum bound, rather than reporting provider-specific reliability measures. If endorsement entities require estimates of measurement reliability for each provider, further work may be needed to develop alternative approaches.

Some readers may object to the idea of reducing the concept of reliability to be a mere function of sample size. Although our reliability calculations under the assumed model are correct, we certainly agree that the beta-binomial approach is unlikely to fully reflect the actual data-generating process. Our goal is not to defend the current framework for estimating reliability, but rather to correct an incoherent (and widely used) formulation. In truth any model of real-world reliability that ignores meaningful covariance is subject to this valid criticism (including the Adams formulation). For example, two identically-sized clusters could differ on estimated reliability within a repeated measures study. Within-cluster variation in quality measures collected across time (indexed by t) or across different quality indicators (indexed by k), such as $\frac{y_{it}}{n_{it}}$ or $\frac{y_{ik}}{n_{ik}}$ might be leveraged to estimate this variance directly.

Such cluster-specific variance estimates could inform reliability studies aggregated over time, patient populations, or types of service. However the beta-binomial model does not easily accommodate multi-dimensional priors. Other approaches to modeling overdispersed or clustered binomial data are available (e.g., mixed-effects logistic or probit models, item response theory), yet such flexibility often comes at the expense of conjugacy. We believe these richer modeling approaches would be a productive direction for future applied work.

The beta-binomial model, unfortunately, has limited access to this richer structure. Each provider's outcome is treated as a single binary response aggregated over an undifferentiated population of patients: there are no repeated measures, no within-provider response patterns, and no patient-level covariates to exploit. Given only (y_i, n_i), sample size is not a proxy for reliability — it is, under the assumed model, reliability's only determinant.

Indeed, the main appeal of the beta-binomial model over other modeling approaches is that the use of the beta distribution as a conjugate prior for π_i enables cluster-specific reliability estimates that are mathematically transparent, easy to calculate by practitioners, and perform stably across a wide-range of cluster sizes and population-level service probabilities. These desirable properties come at the cost of fidelity. In areas of health policy where there is broad agreement that additional features should be incorporated into the measurement model, the beta-binomial will lack the representational capacity to reflect the complexity of quality data. Even in the context of a unidimensional quality measure, non-conjugate priors might better capture meaningful aspects of the data-generating

process, highlighting the importance of the analyst's judgment in selecting the best approach during the estimation process.

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